

POSTS AND TELECOMMUNICATIONS INSTITUTE OF
TECHNOLOGY

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**Chapter 3: LINE AND SURFACE
INTEGRALS**

CALCULUS 2

Faculty of Fundamental Science 1

Hanoi - 2022

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I.Line integral of type 1

1. Definition

Let the function $f(x,y)$ define on a plane arc $\widehat{AB}$.

Divide the arc $\widehat{AB}$ by n points

$$A_0 \equiv A, A_1, \dots, A_{i-1}, A_i, \dots, A_n \equiv B$$

Set

$\Delta s_i, i = \overline{1, n}$ is called the arc length of $A_{i-1}A_i$.

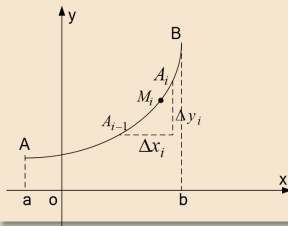
Choose the

arbitrary points $M_i(x_i, y_i) \in A_{i-1}A_i, (i = \overline{1, n})$.

Then, $I_n = \sum_{i=1}^n f(x_i, y_i) \Delta S_i$

is called the sum of first-order

line integrals of the function $f(x, y)$ on arc $\widehat{AB}$.



I. Line integral of type 1

If $n \rightarrow \infty$ so that $\Delta s_i \rightarrow 0$, I_n converges to I regardless of the division of arc $\widehat{AB}$ and the choice of $M_i (x_i, y_i) \in A_{i-1}A_i, (i = \overline{1, n})$, then the number I is called the first-order line integral of $f(x, y)$ along arc $\widehat{AB}$ and the symbol $\int_{\widehat{AB}} f(x, y) ds$.

So $I = \lim_{\max \Delta s_i \rightarrow 0} \sum_{i=1}^n f(x_i, y_i) \Delta S_i = \int_{\widehat{AB}} f(x, y) ds$, where dS denotes the length factor of the arc or the differential of arc.

If the function $f(x, y, z)$ is integrable on arc $\widehat{AB} \subset \mathbb{R}^3$ then the first-order line integral of $f(x, y, z)$ on arc $\widehat{AB}$ denoted is

$$I = \int_{\widehat{AB}} f(x, y, z) ds.$$

I.Line integral of type 1

- The arc $\widehat{AB}$ is called smooth if its tangent is variable continuous.
- The arc $\widehat{AB}$ is called a segmented smooth arc if arc $\widehat{AB}$ can be divided into a finite number of smooth arcs.
- It can be proved: If arc $\widehat{AB}$ is smooth or smooth each segment and $f(x,y)$ is continuous on arc $\widehat{AB}$, then $f(x,y)$ is integrable on arc $\widehat{AB}$.
- The first-order line product has the same properties as the product definite stool

$$\int_{\widehat{AB}} (\alpha f + \beta g)(x,y) ds = \alpha \int_{\widehat{AB}} f(x,y) ds + \beta \int_{\widehat{AB}} g(x,y) ds.$$
$$\int_{\widehat{AC}} f(x,y) ds = \int_{\widehat{AB}} f(x,y) ds + \int_{\widehat{BC}} f(x,y) ds.$$

I. Line integral of type 1

Remark 1

- a) From the above definition, we see the direction of arc $\widehat{AB}$ plays no role because I_n does not depend on the direction of arc AB. So

$$\int_{\widehat{AB}} f(x, y) ds = \int_{\widehat{BA}} f(x, y) ds$$

- b) If l is the length of arc $\widehat{AB}$, then $l = \int_{\widehat{AB}} ds$

I. Line integral of type 1

Remark 1

c) If a material wire has arc $\widehat{AB}$ and mass density is $\rho(x, y)$, then the mass of the material wire is calculated according to the formula

$$m = \int_{\widehat{AB}} \rho(x, y) ds.$$

The center of mass of the wire with density function $\rho(x, y)$ is located at the point $G(x_G, y_G)$, where

$$x_G = \frac{1}{m} \int_{\widehat{AB}} x \rho(x, y) ds, \quad y_G = \frac{1}{m} \int_{\widehat{AB}} y \rho(x, y) ds.$$

I. Line integral of type 1

2. The formula for first-order line integral

a. The arc $\widehat{AB}$ has the general form:

Case 1: Let $\widehat{AB}$ be smooth segmented arc of the form

$y = y(x)$, $x \in [a, b]$ and the function $f(x, y)$ is continuous on $\widehat{AB}$. Then

$$I = \int_{\widehat{AB}} f(x, y) ds = \int_a^b f(x, y(x)) \sqrt{1 + y'^2(x)} dx \quad (3.1)$$

Case 2: Let $\widehat{AB}$ be smooth segmented arc of the form

$x = x(y)$, $y \in [c, d]$ and the function $f(x, y)$ is continuous on $\widehat{AB}$. Then

$$I = \int_{\widehat{AB}} f(x, y) ds = \int_c^d f(x(y), y) \sqrt{1 + x'^2(y)} dy \quad (3.2)$$

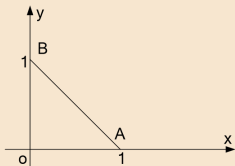
I. Line integral of type 1

Example 1.

Calculate $\int_C (x + y) ds$, where C is the boundary of the triangle with points $O(0, 0)$, $A(1, 0)$, $B(0, 1)$.

Solution

$$\int_C = \int_{\overline{OA}} + \int_{\overline{AB}} + \int_{\overline{BO}}.$$



The segment $\overline{OA}$ has the equation $y = 0, 0 \leq x \leq 1$

$$\int_{\overline{OA}} (x + y) ds = \int_0^1 x \sqrt{1 + 0} dx = \left. \frac{1}{2} x^2 \right|_0^1 = \frac{1}{2}.$$

I. Line integral of type 1

Continuity example 1.

The arc $\widehat{AB}$ has the equation $y = 1 - x, 0 \leq x \leq 1$

$$\Rightarrow \int_{\widehat{AB}} (x + y) ds = \int_0^1 1\sqrt{1+1} dx = \sqrt{2}$$

The segment $\overline{BO}$ has the equation $x = 0, 0 \leq y \leq 1$

$$\int_{\overline{BO}} (x + y) ds = \int_0^1 y\sqrt{1+0} dy = \frac{1}{2}y^2 \Big|_0^1 = \frac{1}{2}$$

$$\Rightarrow \int_C (x + y) ds = 1 + \sqrt{2}.$$

I. Line integral of type 1

2. The formula for first-order line integrals

b. The arc $\widehat{AB}$ has parametric form in the plane

Let $\widehat{AB}$ be smooth segmented arc of the form

$$\widehat{AB} : \begin{cases} x = x(t) \\ y = y(t) \end{cases}, t_1 \leq t \leq t_2$$

and the function $f(x, y)$ is continuous on $\widehat{AB}$. Then

$$I = \int_{\widehat{AB}} f(x, y) ds = \int_{t_1}^{t_2} f[x(t), y(t)] \sqrt{x'^2(t) + y'^2(t)} dt$$

I. Line integral of type 1

Note: The curves in space

$$\widehat{AB} \subset \mathbb{R}^3 : \begin{cases} x = x(t) \\ y = y(t), t_1 \leq t \leq t_2 \\ z = z(t) \end{cases}$$

$$\int_{\widehat{AB}} f(x, y, z) ds = \int_{t_1}^{t_2} f(x(t), y(t), z(t)) \sqrt{x'^2(t) + y'^2(t) + z'^2(t)} dt.$$

c. The curve in polar coordinates

$$\widehat{AB} : r = r(\varphi), \varphi_1 \leq \varphi \leq \varphi_2 \Rightarrow x'^2(\varphi) + y'^2(\varphi) = r^2(\varphi) + r'^2(\varphi)$$

$$I \int_{\widehat{AB}} f(x, y) ds = \int_{\varphi_1}^{\varphi_2} f[r(\varphi) \cos \varphi, r(\varphi) \sin \varphi] \sqrt{r^2(\varphi) + r'^2(\varphi)} d\varphi$$

I.Line integral of type 1

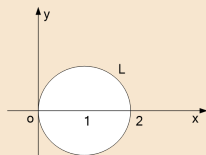
Example 2.

Calculating

$$I = \int_L \sqrt{x^2 + y^2} ds, \text{ where } L \text{ is the circle } x^2 + y^2 = 2x.$$

The equation of the line L in polar coordinates has the form

$$r = 2 \cos \varphi, -\frac{\pi}{2} \leq \varphi \leq \frac{\pi}{2}$$



$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2 \cos \varphi \sqrt{4 \cos^2 \varphi + 4 \sin^2 \varphi} d\varphi = 8 \int_0^{\frac{\pi}{2}} \cos \varphi d\varphi = 8 \sin \varphi \Big|_0^{\frac{\pi}{2}} = 8$$

It is possible to integrate as a parameter $\begin{cases} x = 1 + \cos t \\ y = \sin t \end{cases}, 0 \leq t \leq 2\pi$

$$I = \int_0^{2\pi} \sqrt{(1 + \cos t)^2 + \sin^2 t} dt = \int_0^{2\pi} \sqrt{2 + 2 \cos t} dt = \int_0^{2\pi} \sqrt{4 \cos^2 \frac{t}{2}} dt = 8.$$

II. Line integral of type 2

1. Problem: Calculate the power of the transformed force

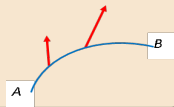
A power produced by force $\vec{F}$ move on the arc L from A to B is

$$W = \vec{F} \cdot \overrightarrow{AB}$$



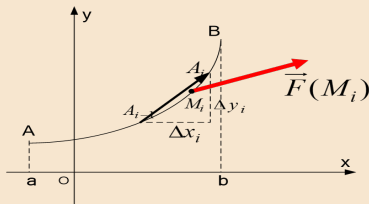
Calculate the power W of the force $\vec{F}(M)$ born while moving on the arc AB from point A to point B

$$\vec{F}(M) = P(M)\vec{i} + Q(M)\vec{j} = (P, Q); M \in AB$$



II. Line integral of type 2

- Divide arc $\widehat{AB}$ by n the points $A_0, A_1, \dots, A_n$. Let the coordinates of the vector $\widehat{A_{i-1}A_i}$ be $\Delta x_i, \Delta y_i$ and the arc length $\widehat{A_{i-1}A_i}$ is $\Delta s_i, i = \overline{1, n}$.



- Choose the arbitrary points $M_i(x_i, y_i) \in \widehat{A_{i-1}A_i}, i = \overline{1, n}$.
- So that the power W of the force produced from A to B on arc AB approximately $W \approx \sum_{i=1}^n P(x_i, y_i) \Delta x_i + Q(x_i, y_i) \Delta y_i$

$$\Rightarrow W = \lim_{\max \Delta s_i \rightarrow 0} \sum_{i=1}^n P(x_i, y_i) \Delta x_i + Q(x_i, y_i) \Delta y_i.$$

II. Line integral of type 2

2. Definition of Line integral of type 2

Given two functions $P(x, y), Q(x, y)$ defined on arc L (or arc $\widehat{AB}$)

- Divide arc L by points

$$A \equiv A_0, A_1, \dots, A_{i-1}, A_i, \dots, A_n \equiv B.$$

- Call the coordinates of the vector $\overrightarrow{A_{i-1}A_i}$ is $\Delta x_i, \Delta y_i$ and arc length $\widehat{A_{i-1}A_i}$ is $\Delta S_i, i = \overline{1, n}$
- Choose the arbitrary points $M_i (x_i, y_i) \in \widehat{A_{i-1}A_i}$.
- Set up totals $I_n = \sum_{i=1}^n P(M_i) \Delta x_i + Q(M_i) \Delta y_i$ it's called the sum of the second-order line segment of the function $P(x, y), Q(x, y)$ along L going from A to B corresponds to a partition of L and a choice $M_i \in \widehat{A_{i-1}A_i}$

II. Line integral of type 2

When $n \rightarrow \infty$ so that $\max \Delta s_i \rightarrow 0$ ($\max \Delta x_i \rightarrow 0, \max \Delta y_i \rightarrow 0$) that I_n converge to a number I regardless of the division of the arc L and the arbitrary choice $M_i \in \widehat{A_{i-1}A_i}$ then the number I is called a line integral of the second type of functions $P(x, y), Q(x, y)$ along arc L from A . Denote by

$$\int_{\widehat{AB}} P(x, y)dx + Q(x, y)dy.$$

Thus

$$I = \int_{\widehat{AB}} P(x, y)dx + Q(x, y)dy = \lim_{\substack{\max \Delta x_i \rightarrow 0 \\ \max \Delta y_i \rightarrow 0}} \sum_{i=1}^n P(x_i, y_i) \Delta x_i + Q(x_i, y_i) \Delta y_i.$$

II. Line integral of type 2

Remark

- Unlike the first-order line integral, in the first-order line integral two, the direction of integration of L is important. If integrating along arc $\widehat{AB}$ going from B to A , the vectors $\overrightarrow{A_{i-1}A_i}$ change direction. So the sum of the integrals will change sign, so

$$\int_{\widehat{AB}} P(x, y)dx + Q(x, y)dy = - \int_{\widehat{BA}} P(x, y)dx + Q(x, y)dy.$$

- The power produced by force $\vec{F} = P(x, y)\vec{i} + Q(x, y)\vec{j}$ for the point to move from A to B along the arc $\widehat{AB}$ will be

$$W = \int_{\widehat{AB}} P(x, y)dx + Q(x, y)dy.$$

II. Line integral of type 2

Remark

- If the $\widehat{AB}$ is a curve in space and the functions $P(x, y, z), Q(x, y, z), R(x, y, z)$ define on the arc $\widehat{AB}$ then product the second-order segmentation of these three functions is also denoted by

$$\int_{\widehat{AB}} P(x, y, z)dx + Q(x, y, z)dy + R(x, y, z)dz.$$

- Let L be a plane curve and closed curve. we use the convention that the positive orientation of a simple closed curve L refers to a single counterclockwise traversal of L . That is a person walking along L in that direction will see the domain bounded by the L closest to me is on the left. The integral taken in the positive direction is denoted by $\oint_L P(x, y)dx + Q(x, y)dy$.

II. Line integral of type 2

Remark

- If two functions $P(x, y), Q(x, y)$ are continuous on smooth arc $\widehat{AB}$ or segmented smooth, then there exists a line integral of the second-order

$$I = \int_{\widehat{AB}} P(x, y)dx + Q(x, y)dy$$

Line integrals of the second-order have the same properties as definite integrals.

Note:

- Sum, difference, multiply a number of the second-order line integrals

$$\int_{\widehat{AC}} Pdx + Qdy = \int_{\widehat{AB}} Pdx + Qdy + \int_{\widehat{BC}} Pdx + Qdy.$$

II. Line integral of type 2

3. The formula for calculating line integrals of type 2

Let the two functions $P(x, y), Q(x, y)$ be continuous on smooth arc $\widehat{AB}$ is given by the parametric equation

$\begin{cases} x = x(t) \\ y = y(t) \end{cases}; A = (x(t_A), y(t_A)), B = (x(t_B), y(t_B)).$ Then

$$I = \int_{t_A}^{t_B} [P(x(t), y(t))x'(t) + Q(x(t), y(t))y'(t)] dt,$$

where arc $\widehat{AB}$ is planar given by an equation of the form

$$y = y(x); A(x_A, y(x_A)), B(x_B, y(x_B))$$

$$I = \int P(x, y)dx + Q(x, y)dy = \int_a^b [P(x, y(x)) + Q(x, y(x))y'(x)] dx.$$

II. Line integral of type 2

Example 1.

Calculating work done by force $\vec{F} = -y\vec{i} + x\vec{j}$ born along the road ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ in its positive orientation.

Solution

Parametric equation of the ellipse

$$\begin{cases} x = a \cos t \\ y = b \sin t \end{cases}, 0 \leq t \leq 2\pi$$

$$A = \int_L xdy - ydx = \int_0^{2\pi} (a \cos t \cdot b \cos t + b \sin t \cdot a \sin t) dt = ab \int_0^{2\pi} dt = 2\pi ab$$

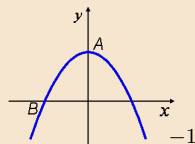
II. Line integral of type 2

Example 2.

Calculating $I = \int_L (2xy - x^2) dx + (x + y^2) dy$

where L is the arc of the parabola

$y = 1 - x^2$ go from point A(0,1) to point B (-1,0)



$$y = 1 - x^2 \Rightarrow dy = -2x dx$$

$$I = \int_0^{-1} [2x(1 - x^2) - x^2 + (x + 1 - 2x^2 + x^4)(-2x)] dx$$

$$= \int_0^{-1} (-2x^5 + 2x^3 - 3x^2) dx$$

$$= \left(-\frac{1}{3}x^6 + \frac{1}{2}x^4 - x^3 \right) \Big|_0^{-1} = -\frac{1}{3} + \frac{1}{2} + 1 = \frac{7}{6}$$

II. Line integral of type 2

3. Green's formula

Theorem 1. (Green's formula)

Let L be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C . If $P(x, y)$ and $Q(x, y)$ have continuous partial derivatives on an open region that contains D , then

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \oint_L P dx + Q dy$$

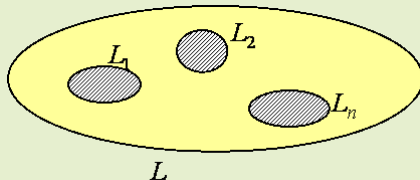
Corollary 1

If D is a simple domain with boundary L , then

$$S_D = \frac{1}{2} \oint_L x dy - y dx = - \oint_L y dx = \oint_L x dy$$

Corollary 2

If D is multidomain, with outer boundary L and inner boundary $L_1, L_2, \dots, L_n$, then



$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \oint_L P dx + Q dy - \sum_{k=1}^n \oint_{L_k} P dx + Q dy$$

II. Line integral of type 2

Example 3.

Evaluate $I = \oint_L (3y - e^{\sin x})dx + (7x + \sqrt{y^4 + 1})dy$, with L is circle $x^2 + y^2 = 4$.

Example 4.

Calculate $J = \int_C (x \arctan x + y^2) dx + (x + 2yx + y^2 e^{-y^3}) dy$, with C is given by the equation $OA : x^2 + y^2 = 2x, y \geq 0$ going from origin to $A(0, 2)$.

II. Line integral of type 2

4. Equivalence propositions for line integrals of type 2

Theorem 2.

Assume that the functions $P(x, y), Q(x, y)$ are continuous with the derivatives their first-order exclusivity in the simple domain D , then the following propositions are equivalent

- $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}, \forall (x, y) \in D$
- $\oint_L Pdx + Qdy = 0$, where L is any closed curve in the domain D .
- The $\int_{\widehat{AB}} Pdx + Qdy$, depends only on 2 points A and B but it does't depend on arc type $\widehat{AB}$ on the domain D .
- the $Pdx + Qdy$ is the total differential of the function some $u(x, y)$ on the domain D .

II. Line integral of type 2

Corollary 1

If $du(x, y) = Pdx + Qdy$ in the D domain, then

$$\int_{\widehat{AB}} Pdx + Qdy = u(B) - u(A)$$

Corollary 2

If $Pdx + Qdy$ is the total differential of the function $u(x, y)$ on the domain $D \in \mathbb{R}^2$ then the function $u(x, y)$ is given by the formula:

$$u(x, y) = \int_{x_0}^x P(x, y)dx + \int_{y_0}^y Q(x_0, y)dy + C \text{ or}$$

$$u(x, y) = \int_{x_0}^x P(x, y_0)dx + \int_{y_0}^y Q(x, y)dy + C, \text{ where}$$

$$M_0(x_0, y_0), M(x, y) \in D.$$

II. Line integral of type 2

Example 5.

Prove that the expression

$$(x^2 - 2xy^2 + 3) dx + (y^2 - 2x^2y + 4y - 5) dy$$

is the total differential of the function $u(x, y)$ on the $\mathbb{R}^2$ and find the function $u(x, y)$.

$$\frac{\partial Q}{\partial x} = -4xy = \frac{\partial P}{\partial y}, \forall (x, y) \in \mathbb{R}^2$$

$$\Rightarrow \exists u(x, y) : \begin{cases} \frac{\partial u}{\partial x} = P(x, y) = x^2 - 2xy^2 + 3 \\ \frac{\partial u}{\partial y} = Q(x, y) = y^2 - 2x^2y + 4y - 5 \end{cases} \Rightarrow \begin{cases} u = \frac{x^3}{3} - x^2y^2 + 3x + \dots \\ \Rightarrow \frac{\partial u}{\partial y} = -2x^2y + 4 - 5 = -2x^2y - 1 \end{cases}$$

$$\Rightarrow f'(y) = y^2 + 4y - 5 \Rightarrow f(y) = \frac{y^3}{3} + 2y^2 - 5y + C$$

$$\Rightarrow u = \frac{x^3}{3} - x^2y^2 + 3x + \frac{y^3}{3} + 2y^2 - 5y + C.$$

II. Line integral of type 2

Example 6.

Evaluate $I = \int_{\widehat{AB}} \frac{xdy - ydx}{x^2 + y^2}$, where $A(1, 1), B(\sqrt{3}, 3)$.

- a) The arc $\widehat{AB}$ is given by the equation: $y = x^2, 1 \leq x \leq \sqrt{3}$.
- b) Let the arc $\widehat{AB}$ make the segment AB a closed curve that does not cover the origin.

III. Surface integral of type 1

1. Definition of surface integral of the of type 1

Let the function $f(M) = f(x, y, z)$ be define on the curved surface S .

- Divide the surface S into n pieces that do not step on each other, name and the symbol for the area, the diameter of the i -th piece is $\Delta S_i, d_i; i = \overline{1, n}$.

- Choose the arbitrary points $M_i (x_i, y_i, z_i) \in \Delta S_i, i = \overline{1, n}$.

- The totals $I_n = \sum_{i=1}^n f(M_i) \Delta S_i$ is called the total surface integral of type one for a division of the surface S and choice of the points $M_i (x_i, y_i, z_i) \in \Delta S_i, i = \overline{1, n}$.

If when $n \rightarrow \infty$ such that $\max d_i \rightarrow 0$ that I_n converges to the number I depends on the division of the surface S and selects the points $M_i \in \Delta S_i$, then the number I is called the first-order surface integral of $f(M)$ on the surface S , denoted by $\iint_S f(x, y, z) dS$.

So
$$I = \iint_S f(x, y, z) dS = \lim_{\max d_i \rightarrow 0} \sum_{j=1}^n f(x_i, y_i, z_i) \Delta S_i.$$

III. Surface integral of type 1

2. Integral conditions and properties of surface integral of type 1.

- If the surface S is smooth (the surface S has a normal variation continuous) or piecewise smooth (dividing S into a finite number of smooth surfaces and the function $f(x, y, z)$ is continuous or piecewise on the surface S , then there exists a first-order surface integral of that function on S .
- A face integral of the first kind has the same properties as a double integral

$$\iint_S (\alpha f + \beta g) dS = \alpha \iint_S f dS + \beta \iint_S g dS.$$

$$\iint_S f dS = \iint_{S_1} f dS + \iint_{S_2} f dS; \quad S = S_1 \cup S_2, S_1 \cap S_2 = \emptyset.$$

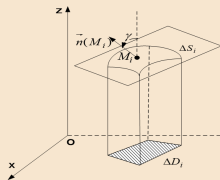
III. Surface integral of type 1

3. Thorem (How to calculate surface integrals of type 1)

Let the function $f(x, y, z)$ be continuity on a smooth surface S given by equation $z = z(x, y)$, $(x, y) \in D$. Then

$$\iint_S f(x, y, z) dS = \iint_D f(x, y, z(x, y)) \sqrt{1 + z'_{x^2}(x, y) + z'_{y^2}(x, y)} dx dy$$

$$0 = dF(x, y, z) = F'_x dx + F'_y dy + F'_z dz$$



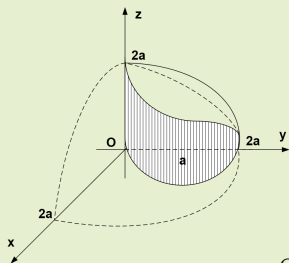
$$\Rightarrow \cos \gamma = \pm \frac{F_z}{\sqrt{F_x'^2 + F_y'^2 + F_z'^2}} = \pm \frac{1}{\sqrt{1 + z_x'^2 + z_y'^2}}$$

$$I_n = \sum_{i=1}^n f(M_i) \Delta S_i \approx \sum_{i=1}^n f(x_i, y_i, z_i) \sqrt{1 + z_x'^2 + z_y'^2} \cdot \Delta D_i; \Delta S_i \approx \frac{\Delta D_i}{|\cos \gamma_i|}.$$

III. Surface integral of type 1

Example 1.

Calculate the area of the upper part of the sphere $x^2 + y^2 + z^2 = 4a^2$ inside the cylinder $x^2 + y^2 \leq 2ay, a > 0$. The upper sphere has the equation $z = \sqrt{4a^2 - x^2 - y^2}$.



$$D : x^2 + (y - a)^2 \leq a^2, x \geq 0$$

$$S = \iint_S dS = \iint_D \sqrt{1 + z_x'^2 + z_y'^2} dx dy$$

$$z'_x = -\frac{x}{z}, z'_y = -\frac{y}{z} \Rightarrow \sqrt{1 + z_x'^2 + z_y'^2} = \frac{2a}{|z|}$$

$$\Rightarrow S = \iint_D \frac{-2a}{\sqrt{4a^2 - x^2 - y^2}} dx dy.$$

III. Surface integral of type 1

Converting to polar coordinates, we get

$$S = 2a \int_0^{\pi} d\varphi \int_0^{2a \sin \varphi} \frac{r dr}{\sqrt{4a^2 - r^2}} = 8a^2 \left(\frac{\pi}{2} - 1 \right).$$

Remark

- The case of surface S is given by the equation $y = y(z, x)$ or $x = x(y, z)$ then we have to project S onto the Ozx or Oyz to find the corresponding double integral.
- In the case of a curved surface of any shape, we must divide it into a number finite parts that satisfy the above theorem, then apply the formula.

III. Surface integral of type 1

4. Applications

- From the definition, we get the formula for surface area curvature S thanks to surface integrals of the first-order surface is $S = \iint_S dS$.
- If S is a material surface, the mass density function is $\rho(x, y, z)$ then the mass of that material surface will be

$$m = \iint_S \rho(x, y, z) dS.$$

- Formula for determining the center of gravity of a curved surface

$$x_G = \frac{1}{m} \iint_S x \rho(M) dS, y_G = \frac{1}{m} \iint_S y \rho(M) dS, z_G = \frac{1}{m} \iint_S z \rho(M) dS.$$

IV. Surface integral of type 2

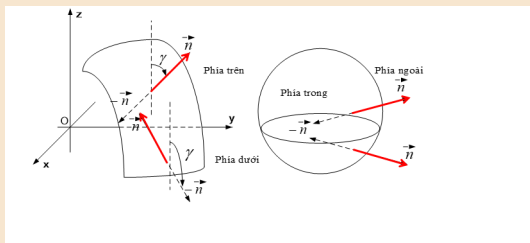
1. Oriented surfaces

- A smooth S -curve is called an oriented if the normal vector unit line $\vec{n}(M)$ completely determined at every $M \in S$ (can subtract the boundary of S) and transform continuously as M runs over S .
- The set $\vec{n}(M), \forall M \in S$ of a oriented curved surface define one side of the surface. Because $-\vec{n}(M)$ is also a normal vector, so the oriented surface always has two sides.

IV. Surface integral of type 2

1. Oriented surfaces

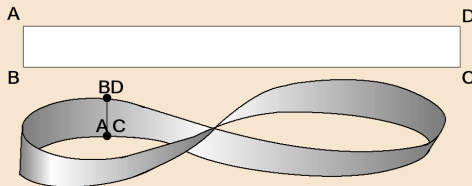
- When the S -curve is not closed and oriented, one usually used from above and below to indicate a specified direction determined by $\vec{n}(M)$. The top of the S -face is the side that $\vec{n}(M)$ with angle Oz axis pointed, and the bottom is the side $\vec{n}(M)$ with Oz axis obtuse angle.
- When the closed S -curve is oriented, one uses the side in and out to describe the specified direction.



IV. Surface integral of type 2

1. Oriented surfaces

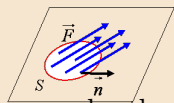
- Outside is the side $\vec{n}(M)$ outward of the object V surrounded by the S -curve, inside is the opposite side.
- There is a curved surface that cannot be oriented, for example, the following surface is called Möbius strip.



IV. Surface integral of type 2

2. Calculate the flux of the vector field through a surface

The Flux of the vector field $\vec{F}$ is constant across the plane



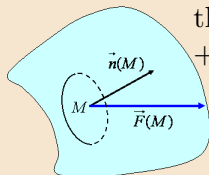
$$\Phi = S \cdot \vec{F} \cdot \vec{n}$$

Divide the surface S into n pieces that do not step on each other, name and the symbol for the area, the diameter of the i -th piece is $\Delta S_i, d_i; i = \overline{1, n}$

+ Choose

the arbitrary points $M_i (x_i, y_i, z_i) \in \Delta S_i, i = \overline{1, n}$.

+ Throughput approx



$$\Phi \approx \Phi_n = \sum_{i=1}^n \Delta S_i \cdot \vec{F}(M_i) \cdot \vec{n}(M_i)$$

IV. Surface integral of type 2

Suppose

$$\vec{F}(M_i) = (P(M_i); Q(M_i); R(M_i)), \vec{n}(M_i) = (\cos \alpha_i; \cos \beta_i; \cos \gamma_i)$$

$$\begin{aligned}\Phi_n &= \sum_{i=1}^n \Delta S_i \vec{F}(M_i) \cdot \vec{n}(M_i) \\ &= \sum_{i=1}^n (P(M_i) \cos \alpha_i + Q(M_i) \cos \beta_i + R(M_i) \cos \gamma_i) \Delta S_i\end{aligned}$$

The flux of the vector field $\vec{F}$ through the S -curve in the direction $\vec{n}$

$$\Phi = \lim_{\max d_i \rightarrow 0} \sum_{i=1}^n (P(M_i) \cos \alpha_i + Q(M_i) \cos \beta_i + R(M_i) \cos \gamma_i) \Delta S_i$$

IV. Surface integral of type 2

3. Definition of the surface integral of type 2

Let the surface S oriented along the normal vector $\vec{n}(M)$ and three functions $P(x, y, z), Q(x, y, z), R(x, y, z)$ determined on S .

- Divide the curved surface S into n pieces that do not step on each other ΔS_i . The symbol for the diameter of the i -th piece is $d_i, i = \overline{1, n}$
- Choose the arbitrary points $M_i (x_i, y_i, z_i) \in \Delta S_i$. The normal vector offace S at point M_i is $\vec{n}(M_i) = (\cos \alpha_i; \cos \beta_i; \cos \gamma_i)$
- Set up totals

$$\begin{aligned} I_n &= \sum_{i=1}^n \Delta S_i \vec{F}(M_i) \cdot \vec{n}(M_i) \\ &= \sum_{i=1}^n [P(M_i) \cos \alpha_i + Q(M_i) \cos \beta_i + R(M_i) \cos \gamma_i] \Delta S_i \end{aligned}$$

IV. Surface integral of type 2

3. Definition of the surface integral of type 2

$$I_n = \sum_{i=1}^n (P(M_i) \cos \alpha_i + Q(M_i) \cos \beta_i + R(M_i) \cos \gamma_i) \Delta S_i$$

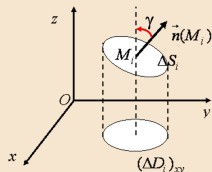
is called the sum of surface integrals of the second type of the three functions P , Q , R taken on the surface S oriented in $\vec{n}(M)$ with one way to divide and one way to choose $M_i \in \Delta S_i, i = 1, \dots, n$.

- If when $n \rightarrow \infty$ so that $\max d_i \rightarrow 0$ but I_n converges to the number I regardless of the division of S and the choice of $M_i \in \Delta S_i$ then the number I is called the face integral of the second kind of the three functions P , Q , R , taken on the surface $I = \iint_S [P(x, y, z) \cos \alpha + Q(x, y, z) \cos \beta + R(x, y, z) \cos \gamma] dS$.
- The integral surfaces of the second type of the vector field $\vec{F}(P, Q, R)$ pass curvature S in the direction $\vec{n}$ is the first-order surface integral of $\vec{F}$.

IV. Surface integral of type 2

3. Definition of the surface integral of type 2

$$I = \iint_S [P(x, y, z) \cos \alpha + Q(x, y, z) \cos \beta + R(x, y, z) \cos \gamma] dS$$



Calling $(\Delta D_i)_{xy}, (\Delta D_i)_{yz}, (\Delta D_i)_{zx}$

is in turn the projection

of ΔS_i onto the coordinate plane Oxy, Oyz, Ozx

$$(\Delta D_i)_{xy} = \Delta S_i \cos \gamma_i \Rightarrow \cos \gamma dS = dxdy$$

$$(\Delta D_i)_{yz} = \Delta S_i \cos \alpha_i \Rightarrow \cos \alpha dS = dydz$$

$$(\Delta D_i)_{zx} = \Delta S_i \cos \beta_i \Rightarrow \cos \beta dS = dx dz$$

Therefore, the face integral of the second kind of the functions P, Q, R on the surface S can sign

$$I = \iint_S P(x, y, z) dydz + Q(x, y, z) dzdx + R(x, y, z) dxdy$$

IV. Surface integral of type 2

Remark 1.

- It has been shown that, if the face S is oriented, smooth or fragmented smooth and the functions P, Q, R , are continuous on S then the second-order surface integral of P, Q, R exists.
- If the direction of the surface integral is changed, then the surface integral of second-order changes sign.
- The surface integrals of the second-order have the same properties as integrals dual.
- The flux of the vector field $\vec{F}(P, Q, R)$ past the curved surface S orientation is calculated by the formula
$$\Phi = \iint_S Pdydz + Qdzdx + Rdx dy.$$
- Assume that the liquid flows through the surface S with velocity $\vec{v}(M)$. Then the flux of the vector field $\vec{v}(M)$ overtaking S is amount of liquid flowing through S in a unit of time.

IV. Surface integral of type 2

a. How to calculate the surface integrals of type 2

Theorem 1.

Suppose $R(x, y, z)$ is continuity on a smooth S-oriented surface for by equation $z = z(x, y)$, $(x, y) \in D \subset (Oxy)$. Then

$$\iint_S R(x, y, z) dx dy = \iint_D R(x, y, z(x, y)) dx dy$$

if the surface integral of the second-order is taken over the surface S

$$\iint_S R(x, y, z) dx dy = - \iint_D R(x, y, z(x, y)) dx dy$$

if the surface integral of the second-order is taken over the surface S .

IV. Surface integral of type 2

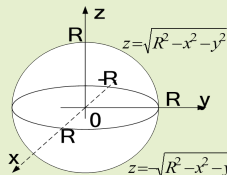
Similarly, we also have

$$\iint_S P(x, y, z) dy dz = \begin{cases} \iint_{D_{yz}} P(x(y, z), y, z) dy dz & \text{when } \cos \alpha \geq 0 \\ - \iint_{D_{yz}} P(x(y, z), y, z) dy dz & \text{when } \cos \alpha \leq 0 \end{cases}$$

$$\iint_S Q(x, y, z) dz dx = \begin{cases} \iint_{D_{zx}} Q(x, y(z, x), z) dz dx & \text{when } \cos \beta \geq 0 \\ - \iint_{D_{zx}} Q(x, y(z, x), z) dz dx & \text{when } \cos \beta \leq 0 \end{cases}$$

IV. Surface integral of type 2

Example 1.



Calculate $I = \iint_S z dx dy$, where

S is the outside of the sphere $x^2 + y^2 + z^2 = R^2$

Divide the sphere into the upper half S_+ and the bottom half S_- there is a way program in turn is $z = \sqrt{R^2 - x^2 - y^2}$ and $z = -\sqrt{R^2 - x^2 - y^2}$

Projecting the halves

of the sphere on Oxy , then we get $D : x^2 + y^2 \leq R^2$

$$I = \iint_{S_+} z dx dy + \iint_{S_-} z dx dy = 2 \iint_D \sqrt{R^2 - x^2 - y^2} dx dy$$

$$I = 2 \int_0^{2\pi} d\varphi \int_0^R \sqrt{R^2 - r^2} r dr = \frac{4}{3} \pi R^3$$

IV. Surface integral of type 2

Example 2.

Find the flux of the vector field over the top of the curved surface $z = x^2 + y^2, -1 \leq x \leq 1, -1 \leq y \leq 1$

$$\Phi = \iint_S z dy dz + x^2 dx dy$$

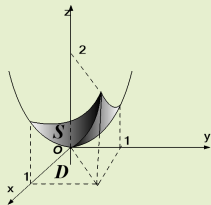
Due to the S -curve against commensurate with the coordinate planes Oyz degrees so

$$\iint_S z dy dz = 0$$

So that

$$\Phi = \iint_S x^2 dx dy = \iint_D x^2 dx dy; \quad D \begin{cases} -1 \leq x \leq 1 \\ -1 \leq y \leq 1 \end{cases}$$

$$\Phi = \iint_D x^2 dx dy = \int_{-1}^1 x^2 dx \int_{-1}^1 dy = \frac{4}{3}.$$



IV. Surface integral of type 2

b. Convert to surface integrals of type 1

Suppose that the $P(x,y,z)$, $Q(x,y,z)$, $R(x,y,z)$ are integrable on the surface S has a normal vector $\vec{n} = (\cos \alpha; \cos \beta; \cos \gamma)$. Then

$$\begin{aligned} I &= \iint_S [P(x, y, z) \cos \alpha + Q(x, y, z) \cos \beta + R(x, y, z) \cos \gamma] dS \\ &= \iint_S \vec{F}(P, Q, R) \cdot \vec{n} \cdot dS \end{aligned}$$

IV. Surface integral of type 2

c. Ostrogradsky- Gauss' formula (O-G)

Let V be a simple solid region and let S be the boundary surface of V , given with positive (outward) orientation. Let P, Q, R be to have continuous partial derivatives on an open region that contains S . Then

$$\iint_S P dydz + Q dzdx + R dxdy = \iiint_V \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dxdydz.$$

Example 4.

Calculate $I = \iint_S x^3 dydz + y^3 dzdx + z^3 dxdy$, where S is the outer surface of the sphere $x^2 + y^2 + z^2 = R^2$.

IV. Surface integral of type 2

Remark 3.

- Consider $P = x, Q = y, R = z$, we get the formula calculate the volume of the body V is $V = \frac{1}{3} \iint_S x dy dz + y dz dx + z dx dy$, where S is oriented outside the domain V .
- It can be considered that the Ostrogradsky-Gauss' formula is extended Green's formula from two-dimensional space to three-dimensional. Sometimes integrating on a non-closed surface S , we can add a curved surface somewhere to apply the Ostrogradsky-Gauss' formula.
- If $\vec{F} = (P, Q, R)$ is a vector field whose component functions have continuous partial derivatives on an open region that contains S , then the Ostrogradsky-Gauss' formula can be written in the form

$$\iint_S \vec{F} \cdot \vec{n} \cdot dS = \iiint_V \operatorname{div} \vec{F} \cdot dx dy dz$$

IV. Surface integral of type 2

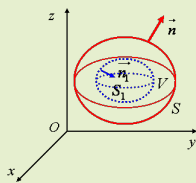
Corollary.

Assume the functions P, Q, R are continuous partial derivatives in the $V \subset \mathbb{R}^3$ whose outer boundary is a closed surface S , the inner boundary is the closed surface S_1 which is smooth each piece. Then

$$\begin{aligned} & \iint_S Pdydz + Qdzdx + Rdx dy - \iint_{S_1} Pdydz + Qdzdx + Rdx dy \\ &= \iiint_V \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dx dy dz. \end{aligned}$$

Especially, if $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 0$, then

$$\iint_S Pdydz + Qdzdx + Rdx dy = \iint_{S_1} Pdydz + Qdzdx + Rdx dy$$



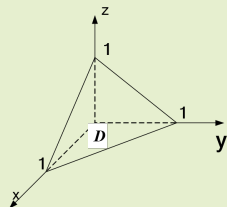
IV. Surface integral of type 2

Example 5.

Calculate $I = \iint_S xz dy dz + yx dz dx + zy dx dy$

take the outside

of the surface S as the boundary of the pyramid



$$x \geq 0, y \geq 0, z \geq 0, x + y + z \leq 1$$

Applying the Ostrogradsky-Gauss' formula

$$I = \iiint_V (z + x + y) dx dy dz$$

$$I = \iint_D dx dy \int_0^{1-x-y} (x + y + z) dz$$

$$\begin{aligned} I &= \iint_D dx dy \left((x + y)z + \frac{1}{2}z^2 \Big|_{z=0}^{1-x-y} \right) = \int_0^1 dx \int_0^{1-x} \frac{1}{2} [1 - (x + y)^2] dy \\ &= \frac{1}{2} \int_0^1 dx \left(y - \frac{1}{3}(x + y)^3 \Big|_{y=0}^{y=1-x} \right) = \frac{1}{2} \int_0^1 \left(1 - x - \frac{1}{3} + \frac{1}{3}x^3 \right) dx = \frac{1}{8} \end{aligned}$$

IV. Surface integral of type 2

Example 6.

Calculating the flux of the vector field $\vec{F} = \frac{q\vec{r}}{r^3}$ across the surface $x^2 + y^2 + z^2 = R^2$ in which q is the charge at the root of the coordinates, $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$.

$$P = q\frac{x}{r^3}, Q = q\frac{y}{r^3}, R = q\frac{z}{r^3}, \forall (x, y, z) \neq (0, 0, 0)$$

Note: We can not apply the Ostrogradsky-Gauss formula in the sphere

$$\Phi = q \iint_S \frac{1}{r^3} (x dy dz + y dz dx + z dx dy)$$

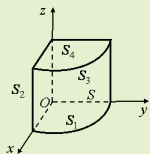
However, the Ostrogradsky-Gauss' formula can be applied to the integral

$$\begin{aligned} \Phi &= \frac{q}{R^3} \iint_S x dy dz + y dz dx + z dx dy \\ &= \frac{q}{R^3} \iiint_V 3 dx dy dz = \frac{q}{R^3} \cdot 3 \cdot \frac{4}{3} \pi R^3 = 4\pi q. \end{aligned}$$

IV. Surface integral of type 2

Example 7.

Calculating the flux of the vector field $\vec{F}(x^3, y^3, z^3)$ through the outside of the cylindrical part $x^2 + y^2 = R^2, x \geq 0, y \geq 0, 0 \leq z \leq h$.



$$\Phi = \iint_S x^3 dydz + y^3 dx dz + z^3 dx dy$$

$$\Phi_k = \iint_{S_k} x^3 dydz + y^3 dx dz + z^3 dx dy, \quad \Phi_1 = \Phi_2 = \Phi_3 = 0$$

$$\Phi_4 = \iint_D h^3 dx dy = h^3 \frac{\pi R^2}{4}; D = \{(x, y) : x^2 + y^2 \leq R^2, x \geq 0, y \geq 0\}.$$

IV. Surface integral of type 2

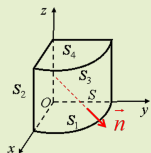
Example 7.

Applying the Ostrogradsky - Gauss' formula, we have

$$\begin{aligned}\Phi + \Phi_1 + \Phi_2 + \Phi_3 + \Phi_4 &= 3 \iiint (x^2 + y^2 + z^2) \, dx dy dz, \\ \iiint_V (x^2 + y^2 + z^2) \, dx dy dz &= \int_0^{\frac{\pi}{2}} d\varphi \int_0^R r dr \int_0^h (r^2 + z^2) \, dz \\ &= \frac{\pi h R^4}{8} + \frac{\pi h^3 R^2}{12} \\ \Phi &= 3 \left(\frac{\pi h R^4}{8} + \frac{\pi h^3 R^2}{12} \right) - h^3 \frac{\pi R^2}{4} = \frac{3\pi h R^4}{8}.\end{aligned}$$

IV. Surface integral of type 2

Example 8.



Calculating the flux of the vector field $\vec{F}(x^3, y^3, z^3)$ through the outside of the cylindrical part $x^2 + y^2 = R^2, x \geq 0, y \geq 0, 0 \leq z \leq h$

We can calculate directly

$$\Phi = \iint_S x^3 dydz + y^3 dx dz + z^3 dx dy$$

$$\vec{n} = \left(\frac{x}{R}, \frac{y}{R}, 0\right) \Rightarrow \iint_S z^3 dx dy = 0$$

$$\iint_S y^3 dx dz = \iint_D \left(\sqrt{R^2 - x^2}\right)^3 dx dz$$

$$\iint_D \left(\sqrt{R^2 - x^2}\right)^3 dx dz = \int_0^R \left(\sqrt{R^2 - x^2}\right)^3 dx \int_0^h dz = h \int_0^R \left(\sqrt{R^2 - x^2}\right)^3 dx$$

IV. Surface integral of type 2

Example 8.

Set $x = R \sin t$, we obtain

$$h \int_0^R \left(\sqrt{R^2 - x^2} \right)^3 dx = hR^4 \int_0^{\pi/2} \cos^4 t dt = \frac{3\pi hR^4}{16}.$$

Thus

$$\iint_S x^3 dydz = \frac{3\pi hR^4}{16} \Rightarrow \Phi = \frac{3\pi hR^4}{8}$$

IV. Surface integral of type 2

d. Stokes' formula

The Stokes' formula extends Green's formula, which is the relationship between the second-order line integral in space and the second-order surface integral.

Theorem 3 (Stokes' theorem)

Assuming that the segmented smooth, oriented S -curve has the boundary of the segmented smooth L and

Let S be an oriented piecewise-smooth surface that is bounded by a simple, closed, piecewise-smooth boundary curve C with positive orientation. Let the functions P, Q , and R be continuous partial derivatives on an open region in $\mathbb{R}^3$ that contains S . Then

$$\oint_C Pdx + Qdy + Rdz =$$

$$\iint_S \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) dydz + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) dzdx + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dxdy.$$

IV. Surface integral of type 2

Remark 2.

- When substituting $z = 0, R(x, y, z) = 0$ into the Stokes formula, we get the Green's formula.

- Let the vector field $\vec{F} = (P, Q, R)$ and

$$\text{rot } \vec{F} = [\vec{\nabla}; \vec{F}] = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right),$$

The Stokes' formula can be written in the form

$$\oint_L Pdx + Qdy + Rdz = \iint_S \text{rot } \vec{F} \cdot \vec{n} \cdot dS = \iint_S \begin{vmatrix} dydz & dx dz & dx dy \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

IV. Surface integral of type 2

Theorem 4 (Theorem of equivalence statements)

Assume the functions P, Q, R are continuous partial derivatives in the simple domain V . Then the following propositions are similar.

- (1) $\frac{\partial R}{\partial y} = \frac{\partial Q}{\partial z}, \frac{\partial P}{\partial z} = \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}, \forall (x, y, z) \in V.$
- (2) $\oint_L Pdx + Qdy + Rdz = 0, L$ is any closed curve in the domain V .
- (3) $\int_{\widehat{AB}} Pdx + Qdy + Rdz$, where $\widehat{AB} \subset V$ doesn't depend on the arc form $\widehat{AB}$.
- (4) Expression $Pdx + Qdy + Rdz$ is the total differential of some function $u(x, y, z)$ on the domain V and

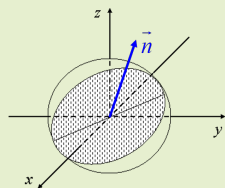
$$\int_{\widehat{AB}} Pdx + Qdy + Rdz = u(B) - u(A)$$

IV. Surface integral of type 2

Example 9.

Calculate $I = \oint_C ydx + zdy + xdz$, where C is the circle, intersection of the sphere $x^2 + y^2 + z^2 = R^2$ and plane $x + y + z = 0$ and direction of C is counterclockwise if looking towards $z > 0$.

Solution



The plane $x + y + z = 0$ passes through the center of the sphere. So the intersection is the great circle. Take the circle as a curved surface S with boundary C . The direction cosines of $\vec{n}$ oriented in the direction of C is $\vec{n} = (1, 1, 1)$. Apply Stokes' formula with $\vec{n}_0 = \frac{1}{\sqrt{3}}(1, 1, 1)$

$$\text{rot } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y & z & x \end{vmatrix} = (-1, -1, -1)$$

$$I = \iint ydx + zdy + xdz = \iint \text{rot } \vec{F} \cdot \vec{n}_0 \cdot dS = -\sqrt{3} \iint dS = -\sqrt{3}\pi R^2.$$